

ATMOS E3.S

AI Capacity Portfolio

Three enterprise deployment modes. One hyperscale neural-data track.

Technical product proposal and conditional business case | v2 | 5 October 2026

Decision lens	Position
Enterprise market entry	Independent appliance, companion service and whole-request capacity pool
Hyperscale opportunity	Neural-Data Sidecar for very large search / recommendation infrastructure
Initial product density	8- and 16-device qualified OEM nodes; 20-22 only after customer and OEM proof
Evidence status	Proposed architecture and planning envelope. No measured ATMOS silicon results or prices supplied.
Economics status	Transparent, independent 36-month scenarios; not quoted prices, forecasts or guaranteed savings

[Open the companion executive presentation](#)

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Reader guide

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1. Portfolio and decision

Lead with SLO-isolated capacity rather than a monolithic heterogeneous inference system. E1, E2 and E3 share one proposed device/runtime/operations foundation, but solve different enterprise adoption and ownership problems. E2 is the first integration priority because a separate service can be piloted without changing the existing GPU runtime. E1 requires a turnkey operations/support distribution; E3 requires the customer's platform and routing integration.

H is a separate design-partner track for hyperscalers and large internet search, social, recommendation and advertising operators. LinkedIn and Meta are market archetypes, not claimed customers or endorsers. A general enterprise search or RAG installation does not establish enough economic or latency pain to justify another accelerator and data lifecycle. Admit an exceptional enterprise only after its scale, memory/network profile and removable cost are demonstrated.

ID / product	Buyer and adoption	Execution boundary	First value proof
E1 / ATMOS AI Appliance	Private AI / neural services; new supported endpoint	Complete, qualified workload on ATMOS	Replace a paid capacity block at identical quality/SLO
E2 / ATMOS Companion Service	Existing GPU estate; independently operated parallel endpoint	Choose GPU OR ATMOS before execution	Remove a forecast GPU overflow purchase or bill
E3 / ATMOS Integrated Capacity Pool	Mature platform team; one gateway and service catalog	Whole-request hardware selection; no token-level crossing	Realize demand placement and measurable bill/lease reduction
H / Hyperscale Neural-Data Sidecar	Hyperscale search / recommendation; sponsored custom integration	Local neural-data operations; compact results to GPU/application	Retire an optimized retrieval tier or avoid its expansion

- Measure accepted work at the same quality, SLO, workload mix, precision policy and availability; capacity alone is not service.
- Keep latency-critical GPU workloads, attention and KV ownership on their existing qualified runtime unless an entire ATMOS workload is independently proven.
- Do not promise native full-LLM execution from HBF capacity, a TT-NN kernel or an OpenAI-compatible HTTP interface.
- Use OEM-qualified four-device domains as a proposed schedulable unit; preserve isolation and include service redundancy.
- Bind releases to a manifest and a named support prime. None of the tracks is a launch commitment before the evidence gates.

1.1 Evidence classes and commercial language

Class	Permitted use	Not permitted
Public component facts [S1-S13]	Choose candidate stacks, controls and qualification questions	Claim ATMOS interoperability, certification or a partner commitment
Private planning inputs	Device/power/topology/model sizing studies	Sell aggregate HBF or TOPS as accepted service
Simulation / emulation	Find bottlenecks, reject infeasible plans, prepare physical tests	Pass a silicon, OEM thermal or customer SLO gate
Matched physical evidence and quotes	Approve the bounded workload, SKU, support and conditional economics	Generalize to different models, platforms or tenants without retesting

2. Common hardware and software foundation

ATMOS is a proposed E3.S accelerator, not a conventional SSD. Every value below comes from the supplied planning envelope; physical silicon, final packaging and an orderable OEM configuration must confirm it. Use decimal GB/TB for HBF rather than silently changing GB to GiB.

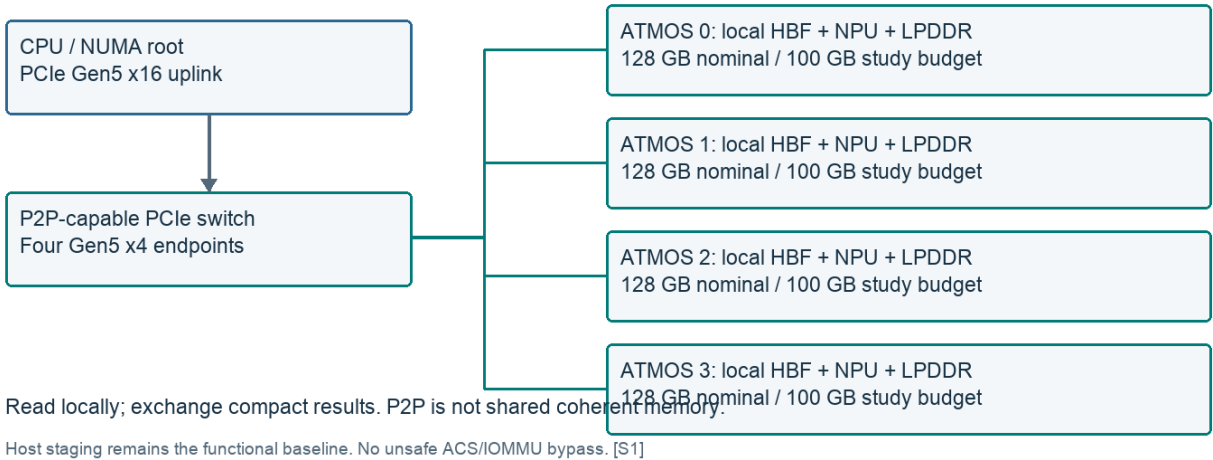
Resource	Planning value	Release condition
Persistent HBF	128 GB nominal/device; 100 GB application study budget	Per-device packing, metadata, replication, scratch and skew must fit
Local HBF read	200 GB/s/device target	Useful sustained rate for actual access pattern; not host/P2P bandwidth
Working memory	8-16 GB LPDDR plus SRAM	Workspace, activations and any whole-model state must fit with reserve
NPU	Approximately 64 TOPS-class source planning target	Define precision, sparsity, clock, kernel coverage and measured service first
PCIe	Gen5 x4 baseline; Gen6 a separate qualification option	Endpoint/backplane/switch/retimer/BIOS lane match and payload proof
Module	E3.S 2T direction; 30-40 W planning envelope	Sustained accelerator-class thermal, power and mechanical certification

Population	Nominal HBF	Application study budget	Device-only power
8 / two 4-device domains	1.024 TB	0.800 TB	240-320 W
16 / four 4-device domains	2.048 TB	1.600 TB	480-640 W
20-22 / stretch only	2.560-2.816 TB	2.000-2.200 TB	600-880 W

All totals are distributed arithmetic sums, not one coherent memory or an application-bandwidth promise. At 8 devices, nominal LPDDR is 64-128 GB distributed; state cannot be pooled without an explicit runtime and communication design. Availability requires complete eligible-service replicas, not just spare memory.

2.1 Four-device PCIe/P2P group

COMMON FOUNDATION / topology to be qualified



Proposed switch-local domain. P2P capability, safe teardown and payload rate require physical proof. [S1]

Gen5 x4 carries approximately 15.75 GB/s per direction before transaction overhead; Gen5 x16 is approximately 63 GB/s. Four x4 endpoints roughly balance an x16 uplink. Eight x4 endpoints behind one x16 uplink introduce about 2:1 aggregate host oversubscription. Two four-device domains use two x16 uplinks, or 32 host lanes: this is a locality/fault-containment choice, not automatic lane savings.

Local HBF processing should dominate transferred bytes. A downstream peer path does not become shared coherent memory, a collective implementation, NVLink, NVSwitch or an AMD scale-up fabric. Cross-root P2P and GPU-to-ATMOS transport are separate platform and cooperating-driver qualifications [S1].

Use host staging as the functional baseline. Enable only documented, supported peer paths with safe buffer ownership, timeout, cancellation and DMA invalidation on reset/removal. Do not bypass IOMMU, ACS, DPC or tenant isolation for a benchmark. P2P is a desired subsystem capability; service economics must still be evaluated with the qualified fallback if P2P is unavailable.

2.2 Software architecture and delivery boundary

Layer	Candidate building block	ATMOS-specific obligation
Northbound service	OpenAI-compatible chat/completions/embeddings where supported	Prove semantics, model catalog and workload envelope; rerank/search are separate API contracts
Gateway	Customer gateway; Agent Router (formerly Envoy AI Gateway) candidate [S12]	Identity, quota, tenant policy, audit, cost metrics and safe endpoint selection
Serving control	KServe or llm-d / Gateway API Inference Extension [S3]	Backend metrics, compatible pool/model contracts, readiness and overload handling
Kubernetes devices	DRA, ResourceSlices, DeviceClasses, ResourceClaims and CDI [S2]	Build/qualify the ATMOS DRA driver; whole-group allocation and tested version matrix
Model runtime	Selected NPU SDK; TT-NN / TT-Metal / vLLM TT plugin candidate [S9]	HBF address/layout, operators, loader, memory planner, batching and complete workload integration
Device plane	Linux PCIe/DMA interfaces [S1]	Driver/firmware/queues/health, reset, safe mapping and peer lifecycle
Release and operations	Container/Kubernetes/OpenTelemetry/Prometheus patterns	Operator, signed ModelPacks, ABI checks, canary/drain/rollback, diagnostics and one support route

The public TT plugin is evidence that vendor integration can use vLLM extension points, not evidence that ATMOS already works [S9]. Pin selected versions and per-model capabilities. ModelPack/ExpertPack is proposed packaging of weights, layout, graph/kernels, metadata and compatibility; a custom 'expertpack' compiler is not an available off-the-shelf ATMOS executable. Weights are data; runtime graph/kernel preparation is a distinct compiler concern.

3. ATMOS AI Appliance

Target buyer: Enterprise / private neural services. Execution boundary: A new, independently operated inference endpoint.

A new private endpoint packaged as an OEM-supported node or small cluster. ATMOS is the primary accelerator only for qualified workloads. Start with embedding, reranking, retrieval or a bounded document-batch operator set; general interactive LLM support is conditional, not implied by device capacity.

Request path: application -> identity/quota gateway -> workload queue and model catalog -> ATMOS model server -> allocated device/group -> result and metering. Streaming or complete responses follow the declared API; batch jobs use explicit job state, retries and cancellation. A GPU fallback, if sold, remains a separate qualified endpoint and is included in both support and cost.

Package	Scope	Qualification focus
Entry / 8 devices	Two four-device domains; 1.024 TB nominal HBF	One frozen model/workload, two active plus reserve service configuration

Package	Scope	Qualification focus
Standard / 16 devices	Four domains; 2.048 TB nominal HBF	Multi-tenant isolation, per-domain fit, sustained power and complete-service HA
Stretch / 20-22	Capacity-driven custom configuration only	Customer need, bay mechanics, x4 lanes, cooling, CPU/NIC/storage and support quote

3.1 Turnkey service contract

Function	Minimum release evidence
Identity / tenant	OIDC or approved API identity, tenant-scoped credentials, audit and DMA isolation
Admission / queues	Context/batch/concurrency caps, bounded queues, overload response and noisy-neighbor tests
Model / API	Signed model/pack digest, tokenizer/precision/operator matrix and golden correctness
Lifecycle	Canary, warmup before readiness, drain, cancel, reset, rollback and controlled replacement
Metering	Accepted/rejected work, end-to-end latency, device time, usage and chargeback; not raw submissions
Availability	Complete replicas and measured failover/restart behavior at required peak demand
Support	One prime, signed BOM and runbook for OEM/device/runtime/control-plane escalation

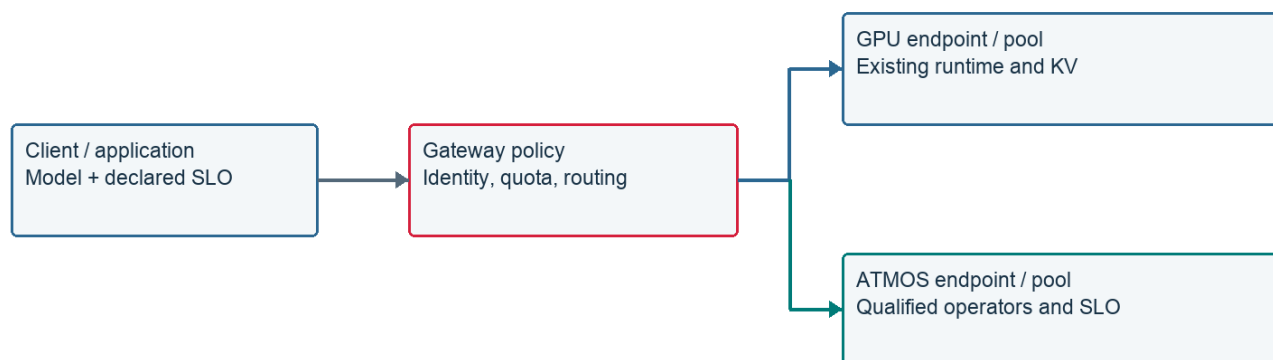
3.2 Efficiency mechanism and cost scenario

The hypothesis is lower fully loaded cost for a capacity-dominant eligible service, not GPU-equivalent single-stream token latency. Validate against the cheapest policy-compliant CPU, GPU or managed API alternative; a small enterprise embedding service may already be cheaper on CPU or pay-per-token APIs.

Section 8 models a replaceable \$180k/year service block, three assumed 8-device nodes including N+1, and \$120k total upfront spend. The hypothetical demand is 100m accepted embedding requests/year with an unchanged input distribution. This is a financial scenario, not a measured rate or a price quote. Exact model, tokens/request, arrival trace, p99, privacy requirement and availability must justify both the comparator and the selected node count.

4. ATMOS Companion Service

Target buyer: Existing enterprise GPU estates. Execution boundary: Choose GPU OR ATMOS before the complete inference request.

ENTERPRISE BOUNDARY / choose one complete-request backend

E2: independent endpoints. E3: unified identity, policy and platform operations.

No in-layer GPU/ATMOS crossing. Shared gateway, CPU and NIC interference still require proof.

E2 and E3 share a complete-request boundary, but differ in gateway and operational ownership.

Deploy an independently operated ATMOS endpoint beside the customer's existing NVIDIA or AMD service. The customer, portal or policy router selects one backend before the request starts. GPU firmware, CUDA/ROCm, model runtime and native collectives stay unchanged. GPU-to-ATMOS peer DMA is not a prerequisite.

Separate endpoint choice avoids a token-level dependency, but does not guarantee unchanged GPU tail latency. Shared CPU, NIC, gateway, storage, power and monitoring can still interfere. Reserve and test these resources under the mixed arrival trace. Reject or queue excess eligible work within the contracted policy rather than silently moving an unqualified workload.

Route to existing GPU service	Route to ATMOS only when qualified
Strict streaming TTFT/TPOT and long-context interactive generation	Embedding, rerank and retrieval service contracts
Dense/VLM workloads or unsupported operators	Document/batch jobs with the declared deadline and model quality
Training/fine-tuning and CUDA/ROCm-dependent work	A complete small/moderate model that passed the operator/state/SLO gate
ATMOS overload/degraded state if compatible fallback is contracted	Healthy ready model/pack and available qualified service capacity

4.1 Deployment and failure contract

- Start with separate endpoints or a parallel cluster; reuse corporate identity and observability without coupling device lifecycles.
- Freeze request semantics, sampling/precision, retry and idempotency rules. A partial stream cannot silently resume on a different model/backend.
- Use health/readiness and admission signals to route new complete requests. An in-flight failure follows the declared restart/fail policy.
- Keep tenant quotas and shared-resource reserves visible. Validate the existing GPU SLO under simultaneous ATMOS saturation and fault injection.

4.2 Efficiency mechanism and cost scenario

Use the companion to avoid an identifiable future GPU overflow block, not to claim a refund for installed GPUs. Section 8 assumes a separately billed \$120k/year capacity block, \$100k upfront ATMOS deployment, and 20m qualified batch jobs/year. Routeable request share is not equal to reclaimable GPU cost: residency, peak demand, batch efficiency

and existing lease terms determine whether the expansion is actually avoided. If only 50% of that budget is removable, this scenario has a negative three-year return.

5. ATMOS Integrated Capacity Pool

Target buyer: Enterprises with platform operations teams. Execution boundary: Whole-request hardware selection behind one qualified gateway.

Expose ATMOS service replicas through the customer's established gateway, identity, quotas, audit and chargeback. The platform chooses a ready, compatible inference pool by model/operator support, workload tier, live queue and measured SLO/unit cost. This is a complete-request resource-pool integration, not an internal Transformer split.

Use separate pools for incompatible model or API contracts. An InferencePool/variant must retain a consistent base model and qualified response semantics [S3]. Do not route merely because two servers accept the same JSON or advertise a similar model name. Precision changes and model versions need explicit quality qualification.

5.1 Kubernetes ownership and constraints

Artifact	Required behavior
ATMOS DRA driver	Publish capacity/topology via ResourceSlices, allocate/configure claims and inject devices with CDI [S2]
DeviceClass / ResourceClaim	Select qualified generation, NUMA/switch domain and workload capability; prevent partial-group conflicts
Atomic group representation	Vendor driver represents a whole qualified group or enforces equivalent atomic placement; DRA does not invent P2P connectivity
ATMOS Operator	Firmware/pack compatibility, warmup readiness, drain, replacement, rollback and fault quarantine
Gateway / endpoint picker	Model/SLO/tier routing, admission, safe fallback and queue/cost/quality/thermal signals
Model readiness	Only register an endpoint after ABI, pack hash, memory fit, correctness and warmup pass
Version / scheduling	Pin kernel/Kubernetes/DRA/API versions and supported features; do not assume DRA preemption or automatic application failover

Example metadata concepts, not a deployable API schema: device-class=atmos-e3s; hbf-capacity=128 GB nominal; application-budget=100 GB study; switch-domain=A; group-size=4; numa=0; qualified-p2p-profile=<manifest-id>; pack-abi=<pinned-version>; workload-class=<tested-embedding/rerank/batch-contract>. The ATMOS driver, naming namespace and claim semantics must be implemented and tested.

5.2 Difference from E2 and proof

Dimension	E2 companion	E3 integrated pool
Front door	Separate service / explicit selection	Unified gateway and catalog
Device/cluster operations	Independent appliance/cluster acceptable	Customer platform, DRA/CDI/operator and release matrix
Routing responsibility	Customer or bounded policy router	Platform endpoint picker and qualified workload policies
Integration burden	Lowest first-pilot scope	Higher multi-tenant, telemetry, admission and lifecycle scope
Execution invariant	One whole-request backend	Same; no in-layer crossing

5.3 Efficiency mechanism and cost scenario

Pooling may improve placement and avoid overbuying, but it does not manufacture GPU savings. Section 8 assumes a \$1m/year estate with \$300k/year of genuinely removable eligible service, \$700k/year retained GPU service, and \$180k upfront ATMOS/platform deployment. The 500m/year request denominator is an unchanged workload mix, not an equivalence between embedding, batch and chat requests. Report full-estate savings separately from the eligible-slice benefit and verify that request routing, cache/residency and peak reserve allow the specific lease or expansion block to be removed.

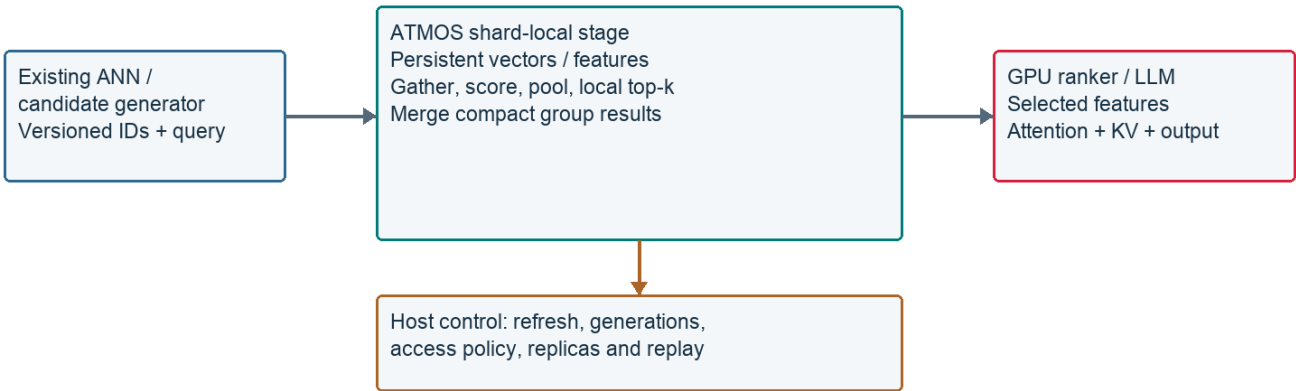
6. Hyperscale Neural-Data Sidecar

Target buyer: Hyperscalers and very large search / recommendation operators. Execution boundary: Shard-local scoring and reduction; compact results to GPU services.

Primary audience: hyperscalers and internet-scale search, social/recommendation, ads, commerce and content platforms with very large persistent vector/feature sets, sustained serving demand, a measured memory/network/latency bottleneck and a custom infrastructure team. Meta's Explore engineering account documents a billions-of-items, multi-stage retrieval/ranking funnel with cacheable embeddings [S4]. The DLRM paper separates embedding-table memory constraints from dense compute [S5]. LinkedIn's LiRank describes production feed, jobs and ads ranking with compression/quantization [S10]. Meta's 2026 sequence-model architecture decouples heavy offline modeling from cached, latency-constrained online ranking [S11]. These are architectural precedents, not evidence of ATMOS adoption or a transferable performance gain. LinkedIn and Meta are target-market examples.

Ordinary enterprise RAG/search is not the default buyer. A small index, low QPS, an optimized CPU/DRAM ANN service, effective caching or a cheap managed endpoint can make the added accelerator and synchronization work uneconomic. An exceptional enterprise must show the same bottleneck and financial evidence as a hyperscaler; a large corpus alone is insufficient.

HYPERSCALE SIDECAR / neural-data stage, not an LLM execution split



Qualification: recall/NDCG, freshness, end-to-end p99, failover and cashable tier removal.
Meta and LinkedIn are target-market archetypes, not customers, partners or validated ATMOS deployments.

H operates at the neural-data stage. The GPU retains its model runtime, attention, KV and generation path.

6.1 Launch scope and ownership

Stage	Owner / contract	Required proof
Candidate generation	Existing ANN/search stack; query plus versioned candidate IDs	Candidate distribution, filters, ACLs and freshness trace frozen
ATMOS local stage	Persistent vector/feature shards; gather, dot/MaxSim, pooling, lightweight scoring and local top-k	Kernel coverage, useful random-read rate, skew and LPDDR workspace

Stage	Owner / contract	Required proof
Group / service merge	Compact IDs/scores/features; deterministic merge and deadline	Fan-out/fan-in, queueing, complete result semantics and failure behavior
GPU/application stage	Existing heavy ranker/LLM/VLM receives qualified compact inputs	No quality, relevance, personalization, safety or end-to-end latency regression
Data lifecycle	Host control plane plus qualified import/refresh and complete service replicas	Atomic index/model generations, replication, deletion/ACL compliance and stale-data rejection

This scope is neural-data processing, not cold-expert/per-layer GPU offload. Keeping compact results at an application boundary preserves a testable product contract. Semantic-cache reuse requires version/ACL/expiry correctness; a cache hit is not assumed safe. Token-level late interaction is a candidate workload only after actual token-vector expansion, MaxSim operators and relevance quality pass. A simple vector dot product is not automatically equivalent to a cross-encoder reranker or a full recommendation model.

6.2 Transparent sizing and traffic example

Hypothetical input / derivation	Result / interpretation
500m vectors x 768 dimensions x 2 bytes	768 GB raw vectors; decimal units
25% hypothetical packing/IDs/metadata allowance	960 GB per generation; actual index/layout metadata must be measured
One 16-device node x 100 GB application study budget	1,600 GB distributed budget; one generation can fit only if every shard fits
Two generations x two complete replicas	3,840 GB data budget; four 16-device nodes / 64 devices assumed for live plus staging replicas
20,000 candidate vectors x 768 x 2 bytes/query	30.72 MB candidate-vector payload if the baseline actually transfers these vectors
200 selected items x (8-byte ID + 4-byte score)	2.4 kB result payload; excludes query/candidate IDs, features, headers, refresh and replicas
20,000 candidates -> 200 heavy-ranker candidates	99% fewer downstream candidates only if the same quality/relevance is retained; not 99% total cost saving
10,000 queries/s x 30.72 MB local read/query	307.2 GB/s useful local vector demand before metadata/cache effects; not an achieved rate

The sizing illustration uses two complete live replicas plus two full staging replicas for a controlled generation switch. A surviving live replica must still meet the peak SLA, or more capacity is required. Four independent PCIe domains in a node do not each contain a complete index. HBF write/import bandwidth, endurance, delete handling and update cost are unproven and may change this node count. Host ANN metadata, journals, CPU DRAM, SSDs and NICs are separately budgeted where they are not included in the HBF layout.

The apparent traffic reduction is valid only for the described vector-moving baseline. An ANN service already returning compact IDs may see little or no inter-stage byte reduction. Measure bytes on the actual bottleneck, including candidate input IDs, scatter/gather, selected features, group merge and refresh. Do not full-scan 768 GB for every query merely because HBF can hold the corpus.

A critical-path illustration, not a p99 prediction: if the removable stage occupies 35 ms of a 100 ms serial budget and is twice as fast, the budget becomes 82.5 ms, or 17.5% lower. End-to-end p99, overlapping work, queueing and fan-out tails must be measured; percentile latencies cannot be added as though they were deterministic components.

6.3 Hyperscale admission and rejection charter

- Sponsor provides index/model size, precision, candidate histogram, QPS/arrival trace, p95/p99 budget, relevance/freshness/ACL requirements and removable spend.
- Benchmark the best optimized CPU/DRAM ANN + cache/compression/SSD tier, GPU-resident variant and incumbent stack; do not choose a weak GPU-only strawman.

- Freeze recall@k/NDCG or recommendation quality, filters, freshness, update rate and complete-service availability before measuring throughput.
- Replay updates, hot/skewed shards, tenant contention, a lost node/group, stale generations and restart while measuring the complete pipeline.
- Proceed only if measured local useful work outweighs additional DMA/RPC/merge, the full result meets p99, and support/replica/refresh costs leave positive unit economics.
- Reject if the data fits cheaply in the incumbent tier, the baseline already sends compact results, relevance drops, updates cannot meet freshness, or no capacity block can be retired/avoided.

Section 8's H scenario is an assumed \$600k/year replaceable retrieval tier, \$900k/year retained GPU service and \$360k upfront deployment of four assumed 16-device nodes. No GPU generation savings or monetized relevance/revenue uplift is included. At only 50% realized tier reduction, the three-year return is negative.

7. Workload fit and matched-service measurement

Workload	Track / control	Required acceptance metric	Position
Embeddings / rerank	E1/E2/E3; CPU/managed/GPU control	Model quality, input length, batch, accepted rate and p99	First bounded service candidates
Document / asynchronous batch	E1/E2/E3; incumbent runtime	Accuracy, completion deadline, cancellation and energy/job	Conditional operator coverage
Complete small/moderate LLM	E1/E2/E3 only after full model proof	TTFT, TPOT/ITL, context, concurrency, sampling and state fit	Not implied by HBF/NPU capacity
Vector/feature refinement / late interaction	H; best CPU/DRAM ANN and GPU variants	Recall/NDCG, freshness, p99, update load and cost/query	Hyperscale design-partner track
Long-context interactive / strict token SLO / VLM	Existing qualified NVIDIA/AMD service	Matched end-to-end and per-token tails	GPU preference unless whole ATMOS service independently passes
Training, fine-tuning and token-synchronous heterogeneous execution	Existing GPU platform / outside scope	No claimed ATMOS product result	Not portfolio launch workloads

7.1 Freeze a workload card before the POC

Field	Required value
Workload identity	Immutable model/dataset/index/tokenizer/pack hashes; licenses and tenant/ACL policy
Arrival and input	Length/candidate/update distributions, batching, concurrency, burst/skew and peak demand
Quality	Golden output / accepted domain score / recall@k or NDCG threshold against the same reference
Service	Sponsor-owned p50/p95/p99, TTFT/TPOT where relevant, deadline, timeout and overload budget
Availability	Complete replica policy, N+1 peak service, restart/failover semantics and maintenance
Comparator	Cheapest qualified policy-compliant incumbent/CPU/managed/GPU configuration and approved contract price
Cash proof	Named removable bill/lease/expansion block, realized fraction, approved quotes and 36-month horizon

7.2 Measurement procedure

1. Freeze manifests and the above workload card; warm up both paths and record actual software/hardware versions.
2. Run identical steady, burst, hot-shard and mixed-tenant traces. Vary offered load until each qualified service reaches its SLO-limited capacity.
3. Count only accepted, quality-correct work that meets the declared SLO. Retain all rejects, errors, timeouts and stale results in the denominator/report.
4. Correlate queue, host, device, DMA, merge, runtime and GPU traces. Record useful HBF bytes, peer/host/NIC bytes, CPU/LPDDR, thermal and facility power.
5. Inject cancellation, module/group/node faults, reboot, pack/version mismatch and data refresh; demonstrate isolation and the contracted failure policy.
6. Apply measured rates and reserve to actual node counts. Obtain OEM/support quotes and verify which baseline spend is truly avoided before recomputing ROI.

Use current qualified baselines, not an old GPU memory shortage alone. H200's published 141 GB HBM3e / 4.8 TB/s per GPU [S7] and GB300 NVL72's 72 GPUs, rounded 20 TB GPU memory and 130 TB/s rack NVLink [S8] show how high-end controls can remove the simple capacity-fit thesis. A rack and an 8/16-device node are not equal service units. Match the workload, service envelope, facility and total contracted cost; GPU memory/BW totals do not imply ATMOS-accessible memory.

8. Cost, efficiency and conditional ROI

ILLUSTRATIVE USD CASH MODEL, NOT A QUOTE OR BENCHMARK. Every device price, service budget, accepted throughput, node count and wattage below is an explicit hypothetical input. Replace them with customer demand, measured whole-service performance and binding OEM/support quotes before investment approval. Scenarios are independent and must not be added together. No ATMOS speedup is assumed.

8.1 Method and financial boundary

Use an undiscounted 36-month cash horizon in constant USD. Baseline avoidable budgets are recurring cloud/lease/service commitments that can actually be cancelled or not purchased; they already include their provider's power. Retained GPU costs are identical on both sides. Existing sunk GPU capital is not reclaimed, and freed capacity is only a noncash benefit unless it avoids a specific future purchase or bill. Taxes, financing, residual value and vendor silicon R&D are not separately modeled; customer procurement quotes must include supplier margin and any allocated program NRE.

Metric	Calculation
Initial investment I	Quoted hardware/system/spares + one-time integration and validation
Annual ATMOS cost A	Support + software + operations + measured facility-adjusted electricity
Annual electricity	Average IT kW x 8,760 hours x PUE 1.4 x \$0.12/kWh; PUE and tariff are assumptions
Three-year net benefit	3 x (avoidable annual baseline x realized fraction - A) - I
Three-year ROI	Three-year net benefit / I; not a vendor margin or measured product return
Simple payback	12 x I / (avoidable annual baseline - A), if annual net saving is positive
Break-even monetization	(I / 3 + A) / avoidable annual baseline for a 36-month investment
Unit economics	Three-year cash cost / three-year accepted service units at unchanged quality, demand, SLO and availability

8.2 Worked scenario results

Track	Initial	ATMOS/year	3y baseline	3y project	Net benefit	TCO saving
E1	\$120,000	\$47,208	\$540,000	\$261,623	\$278,377	51.6%
E2	\$100,000	\$43,766	\$360,000	\$231,298	\$128,702	35.8%
E3	\$180,000	\$82,943	\$3,000,000	\$2,528,830	\$471,170	15.7%
H	\$360,000	\$188,830	\$4,500,000	\$3,626,490	\$873,510	19.4%

Track	3y ROI	Payback	Baseline / new per 1,000 units	Annual accepted demand
E1	232.0%	10.8 months	\$1.8000 / \$0.8721	100,000,000 embedding requests
E2	128.7%	15.7 months	\$6.0000 / \$3.8550	20,000,000 qualified batch jobs
E3	261.8%	10.0 months	\$2.0000 / \$1.6859	500,000,000 requests in the unchanged estate workload mix

Track	3y ROI	Payback	Baseline / new per 1,000 units	Annual accepted demand
H	242.6%	10.5 months	\$0.1500 / \$0.1209	10,000,000,000 search/recommendation queries

E3's headline percentage applies to the full \$1m/year estate, including \$700k/year retained GPU services. H includes \$900k/year retained GPU services. The E3 denominator uses the identical, frozen workload mix; it is not a claim that different request types have equal cost. Service units differ by track and are not cross-comparable.

8.3 Inputs, configuration and realization risk

E1: scenario ledger

Baseline: A cancellable cloud/leased embedding-service block costing \$180k/year.

Configuration: Three assumed \$30k 8-device nodes; two active plus one N+1 node.

Input	Hypothetical budget
Hardware and included reserve	\$90,000
One-time rollout, migration, qualification and integration	\$30,000
Annual support / software / operations	\$12,000 / \$8,000 / \$25,000
Average complete-block IT power / annual facility electricity	1.5 kW / \$2,208
Annual avoidable baseline / retained service	\$180,000 / \$0

36-month break-even requires more than 48.4% of the specified baseline spend to be realized as avoided bills. If only half is monetized, net three-year benefit becomes \$8,377. Revenue growth, latency improvement and capacity credit are excluded from this cash return.

E2: scenario ledger

Baseline: An otherwise required, separately billed \$120k/year GPU overflow block.

Configuration: Three assumed \$25k 8-device nodes; two active plus one N+1 node.

Input	Hypothetical budget
Hardware and included reserve	\$75,000
One-time rollout, migration, qualification and integration	\$25,000
Annual support / software / operations	\$10,000 / \$8,000 / \$24,000
Average complete-block IT power / annual facility electricity	1.2 kW / \$1,766
Annual avoidable baseline / retained service	\$120,000 / \$0

36-month break-even requires more than 64.2% of the specified baseline spend to be realized as avoided bills. If only half is monetized, net three-year benefit becomes -\$51,298. Revenue growth, latency improvement and capacity credit are excluded from this cash return.

E3: scenario ledger

Baseline: A \$1m/year estate with a \$300k/year eligible, genuinely removable capacity block.

Configuration: Three assumed \$40k 16-device nodes; two active plus one N+1 node.

Input	Hypothetical budget
Hardware and included reserve	\$120,000
One-time rollout, migration, qualification and integration	\$60,000
Annual support / software / operations	\$18,000 / \$12,000 / \$50,000
Average complete-block IT power / annual facility electricity	2 kW / \$2,943

Input	Hypothetical budget
Annual avoidable baseline / retained service	\$300,000 / \$700,000

36-month break-even requires more than 47.6% of the specified baseline spend to be realized as avoided bills. If only half is monetized, net three-year benefit becomes \$21,170. Revenue growth, latency improvement and capacity credit are excluded from this cash return.

H: scenario ledger

Baseline: A \$600k/year replaceable optimized retrieval tier plus \$900k/year retained GPU services.

Configuration: Four assumed \$60k 16-device nodes: two index generations times two service replicas.

Input	Hypothetical budget
Hardware and included reserve	\$240,000
One-time rollout, migration, qualification and integration	\$120,000
Annual support / software / operations	\$36,000 / \$24,000 / \$120,000
Average complete-block IT power / annual facility electricity	6 kW / \$8,830
Annual avoidable baseline / retained service	\$600,000 / \$900,000

36-month break-even requires more than 51.5% of the specified baseline spend to be realized as avoided bills. If only half is monetized, net three-year benefit becomes -\$26,490. Revenue growth, latency improvement and capacity credit are excluded from this cash return.

8.4 Sensitivity and stop rule

Track	0% realized	50% realized	75% realized	100% realized	Break-even floor
E1	-\$261,623	\$8,377	\$143,377	\$278,377	48.4%
E2	-\$231,298	-\$51,298	\$38,702	\$128,702	64.2%
E3	-\$428,830	\$21,170	\$246,170	\$471,170	47.6%
H	-\$926,490	-\$26,490	\$423,510	\$873,510	51.5%

Fail the economic gate if the matched-SLO measured node count increases, qualified capacity or fleet demand is insufficient, baseline leases cannot be reduced, or index refresh, replicas, CPU/NIC load, OEM NRE and support erase the margin. Recalculate for a 25% capex increase, 25% lower goodput, higher support cost and the customer's tariff/PUE; do not substitute nominal HBF bandwidth or TOPS for accepted service. Directly compare H with optimized CPU/DRAM plus ANN, compression, caching and SSD-tiering alternatives as well as GPU-resident implementations.

8.5 Published-price counterexample: a managed API can win

Public reference / hypothetical demand	Input-token charge
text-embedding-3-small: \$0.02 per million input tokens [S13]	\$1,024/year at 100m requests x 512 tokens
text-embedding-3-large: \$0.13 per million input tokens [S13]	\$6,656/year at the same token count

These published standard prices were reviewed on 5 October 2026 and can change. They are input-token charges only, not full retrieval/storage/network/support costs or a private deployment quote. The model/embedding dimension and quality may differ. If a permitted managed API meets the same privacy, quality, data residency, availability and latency contract, E1's assumed \$180k/year private baseline is not the right alternative and its positive example does not apply. The small-model token charge is \$3,072 over three years versus E1's \$261,623 hypothetical deployment cost before any API-side additions. Do not justify an appliance for ordinary embedding calls using an artificially expensive baseline.

9. OEM, thermal, availability and support

Bay count is not accelerator support. The verified Supermicro ASG-2115S-NE332R example has 32 front E3.S 1T Gen5 x2 storage bays [S6]; that is not a drop-in match for the proposed x4, 2T, 30-40 W ATMOS module. Seek named,

orderable OEM configurations and a signed accelerator-class qualification, not a percentage of a storage bay population. Dell/HPE and other OEM options require the same BOM/topology/thermal proof; no unsupported compatibility assertion is made here.

Stage	Test scope	Advance evidence
1 module	Power, useful HBF/NPU service, inlet/throttle, voltage and reset	Sustained qualified service and mechanical/electrical envelope
4-module group	Adjacent heat, switch/retimer load, P2P, security and removal	Safe lifecycle, lane map and complete host-staged/peer-path comparison
8-module node	NUMA, CPU/NIC load, multi-tenant saturation, full soak and one-domain failure	Accepted service, fault containment and complete replica policy
16-module standard	Maximum inlet, N+1 PSU/fan, maintenance, spares and support	Orderable BOM, full service under failure/maintenance and cost quote
20-22 stretch	Customer-specific fit, all bays, NIC/storage reserve and cooling	Only after standard SKU proof; no initial launch promise

16-device power item	Budget treatment
Devices	480-640 W planning arithmetic; measure sustained workload/idle/peak
CPU / DRAM	OEM-specific; source planning 300-700+ W is not a certified BOM
Switch / retimer / backplane / NIC / SSD	Measure rather than omit; maintain lane and thermal headroom
Fans / conversion / facility	Wall power, N+1/inlet condition and customer PUE; module wattage is not system energy

9.1 Required configuration contract

- Mechanics: exact E3.S thickness/carrier, bay spacing, retention, connector rating, service access and hot-replacement rules.
- Electrical: root/switch/NUMA map, actual x4/Gen5 or separately qualified Gen6, BAR/IOMMU, AER/DPC/FLR and safe peer-path enablement.
- Thermal: continuous qualified workload at maximum inlet, adjacent modules, one-fan failure, connector temperatures and throttling behavior.
- Firmware/security: signed BIOS/BMC/device updates, rollback compatibility, health telemetry, DMA isolation and failure containment.
- Availability: complete model/index service replicas with enough surviving peak capacity; disjoint unique shards are not HA.
- Commercial: all-in BOM, NRE, device/switch/runtime support, lead times, spares, geography, escalation and liability boundary.

9.2 One accountable support route

Layer	Accountable owner / required interface
Chassis, power/cooling, BIOS/BMC	OEM/integrator; named SKU and field-service SLA
Switch/backplane and P2P	OEM + switch vendor + SanDisk; one root-cause/firmware route
ATMOS module, firmware, SDK/driver	SanDisk; diagnostics, reset/isolation and replacement contract
NPU compiler/runtime/operators	SanDisk + selected NPU partner; signed scope and version/test matrix

Layer	Accountable owner / required interface
DRA/operator/serving adapters	SanDisk distribution owner; readiness, drain/rollback and issue response
Customer gateway/data pipeline	Named customer/software partner; routing, refresh, ACL and support boundary
First call / economic case	One prime per released package; support included in approved ROI ledger

10. Software deliverability and release package

Workstream	Existing foundation	Build or qualify before release
Device/SDK	Linux DMA/PCIe and selected NPU APIs	ATMOS queue ABI, safe mappings, HBF/LPDDR planner, health and recovery
Workload runtime	Candidate TT-NN/TT-Metal/vLLM plugin [S9]	Actual ATMOS operators, graphs/kernels, model loader, batching and memory/service curve
Packs	Container/signing/model artifact patterns	ModelPack/feature-pack manifest, precision/layout, immutable hashes, verification and rollback
Kubernetes	DRA/CDI and standard controllers [S2]	ATMOS DRA driver, atomic group policy, operator, readiness/drain and taint/quarantine
Serving/gateway	KServe/llm-d/customer gateway patterns [S3]	Qualified API/model semantics, endpoint metrics, route/admission and tenant contracts
Hyperscale data	Incumbent ANN/feature pipeline	Local kernels, shard/merge, generation protocol, update/delete/ACL and complete replicas
Evidence/support	OpenTelemetry/Prometheus/CI conventions	Correlated traces, golden suites, fault harness, signed BOM, support/runbooks and cost worksheet

10.1 Minimum compatibility manifest

- Hardware/device firmware/NPU generation; switch, retimer, backplane, carrier and per-root lane topology.
- OEM, CPU/NUMA, BIOS/BMC, inlet/power policy, NIC/SSD/host-memory budget and qualified security controls.
- OS/kernel, driver/SDK/runtime, containers, Kubernetes/DRA/CDI/operator and gateway/serving versions.
- Model/tokenizer/pack/index hashes, operator/precision/layout, per-device memory fit, supported API and workload envelope.
- Peak arrival/load, quality and freshness thresholds, SLO, timeout/cancel/retry policy, replica/reserve and failure response.
- Quotes and cost inputs, known limits, first-call support, replacement/rollback procedure and evidence bundle IDs.

Pin actual upstream releases; neither a DRA object nor a model family in a vendor README proves the ATMOS device path. Do not expose a pod as ready until device checks, ABI/pack match, memory plan, golden correctness and warmup pass. Treat security and update/delete semantics as release gates, not post-pilot cleanup.

11. Gated proof and adoption program

The gates are evidence order, not a promised silicon or launch schedule. Simulation/emulation can refine or reject the plan while hardware is unavailable, but cannot replace the physical service, thermal, support or customer-economic gates.

Gate	Scope / owner	Required exit package
G0 / workload and economics	Product + customer sponsor + finance	One immutable workload card, best comparator, bill/expansion block, SLO/quality, baseline traces and quote plan
G1 / one device	SanDisk + NPU partner	Operators/correctness, HBF useful read, LPDDR fit, power/thermal, queue/DMA/reset and service curve
G2 / four-device domain	SanDisk + OEM/switch vendor	Host/peer comparison, fan-out/merge, isolation, safe teardown, skew and domain-fault evidence
G3 / 8 then 16-device appliance	SanDisk distribution + OEM/support prime	API/admission, tenant interference, full soak, complete replicas/peak failover, signed BOM/support
G4 / E2 or E3 customer integration	Customer platform + SanDisk	Whole-request routing, GPU tails protected, ready/drain/rollback, measured costs and approved removable capacity
G5 / H design partner	Hyperscale search/data sponsor + SanDisk	Optimized incumbent comparison, recall/NDCG/freshness, refresh/delete/ACL, full-pipeline p99 and actual tier avoidance

11.1 Smallest useful first pilots

- Enterprise: one existing paid workload and one ATMOS endpoint using E2. Do not expand to every model/operator or a unified gateway before matched service is demonstrated.
- Appliance: turn that proven workload into an OEM/runbook/support package, then qualify the 8/16-device product envelope.
- Integrated pool: add E3 only for a customer with DRA/operator/gateway ownership and a measurable placement/removable-cost problem.
- Hyperscale: one neural-data operator stage, one trace and one index-generation protocol. No assumption that vector scoring, late interaction and recommendations all share one kernel contract.

11.2 Stop, narrow or return to baseline

- Stop on incorrect outputs, relevance/freshness/ACL regression, unsafe DMA/reset or a missing required operator.
- Narrow when lower-level evidence passes but complete service fails its p99, queue, context, concurrency or availability envelope.
- Use the qualified staged path or redesign the OEM fixture if peer access is unsupported; do not relax security to keep a claim.
- Reject density or revise node counts if thermal derating, shared CPU/NIC load, replicas, staging or repair reserve changes accepted capacity.
- Fail the business gate if the realized fraction falls below Section 8's break-even floor, quotes/NRE rise, or incumbent optimization wins.

12. Investment decision and customer discovery

Decision	Recommendation / required next evidence
Enterprise product charter	Approve E1/E2/E3 definitions; lead one E2 matched-service pilot and a supported 8-device path
OEM scope	Approve a four-device fixture and 8->16 qualification; hold 20-22 until standard-node proof
Software ownership	Name driver/runtime/operator/gateway/support owners and signed NPU/OEM deliverables
Hyperscale charter	Sponsor H only with large-search bottleneck, update/quality/SLO trace and an actually removable retrieval tier
Commercial claims	Hold all device specifications, throughput, efficiency, savings and launch dates pending matched physical evidence and quotes
Finance gate	Publish complete cash and noncash benefits separately; approve only the recomputed, bounded business case

12.1 Discovery questions that decide fit

- Which exact service, model/index and bill or future capacity purchase could ATMOS replace? Is that spend contractually avoidable?
- What are current accepted throughput, peak arrival, p99/TTFT/TPOT, quality and freshness? Which stage actually limits the service?
- How large is the persistent data/model and working state per shard? What is the measured byte movement, cache behavior and update/delete rate?
- What is the cheapest qualified CPU/managed/GPU alternative after normal caching, batching, ANN/compression and routing improvements?
- Who owns API/model semantics, Kubernetes/DRA/gateway, data refresh, security, OEM lifecycle and first-call support?
- What complete replica and peak reserve survives a module/domain/node fault, and what change would invalidate the cost model?

Recommended proof sequence: matched baseline -> one operator/device -> qualified group -> supported whole service -> customer routing and actual avoided spend. Make H a separate hyperscale workload and data-lifecycle charter rather than a generic enterprise add-on.

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Appendix A. Public research and claim boundaries

Primary public references below were reviewed for architectural constraints and market context. Dates/specifications are source-specific. They do not validate ATMOS hardware, customers, partnership, prices or measured performance. Private device values originate in the supplied planning proposal; economic inputs in Section 8 are explicitly hypothetical. No unresolved search-result identifiers are used as citations.

S1. Linux kernel: PCI Peer-to-Peer DMA Support

<https://docs.kernel.org/driver-api/pci/p2pdma.html>

Basis: Topology limits, cooperating provider/client/orchestrator drivers, safe DMA invalidation. Not ATMOS qualification. Reviewed 5 October 2026.

S2. Kubernetes: Dynamic Resource Allocation

<https://kubernetes.io/docs/concepts/scheduling-eviction/dynamic-resource-allocation/>

Basis: ResourceSlices, DeviceClasses, ResourceClaims and CDI. Version and feature status must be pinned; DRA is not model readiness or failover. Reviewed 5 October 2026.

S3. llm-d: Architecture

<https://llm-d.ai/docs/architecture>

Basis: Router, endpoint picker, InferencePool, model servers and workload variants. ATMOS still requires its own backend, metrics and qualification. Reviewed 5 October 2026.

S4. Meta: Scaling the Instagram Explore recommendations system (9 Aug 2023)

<https://engineering.fb.com/2023/08/09/ml-applications/scaling-instagram-explore-recommendations-system/>

Basis: Billions of items; cached two-tower retrieval; multi-stage ranking that narrows thousands of candidates. Market-scale precedent, not ATMOS adoption. Reviewed 5 October 2026.

S5. Meta authors: Deep Learning Recommendation Model (31 May 2019)

<https://arxiv.org/abs/1906.00091>

Basis: Embedding-table memory constraints and separate dense compute. Older architecture evidence, not a current measured ATMOS business case. Reviewed 5 October 2026.

S6. Supermicro: ASG-2115S-NE332R product specifications

<https://www.supermicro.com/en/products/system/storage/2u/asg-2115s-ne332r>

Basis: 32 front E3.S 1T Gen5 x2 storage bays. Demonstrates why bay count is not proof of x4, 2T, accelerator-class support. Reviewed 5 October 2026.

S7. NVIDIA: H200 product specifications

<https://www.nvidia.com/en-us/data-center/h200/>

Basis: 141 GB HBM3e and 4.8 TB/s per GPU; distinct SXM/NVL configurations. A strong installed-base control, not a price quote. Reviewed 5 October 2026.

S8. NVIDIA: GB300 NVL72 product specifications

<https://www.nvidia.com/en-us/data-center/gb300-nvl72/>

Basis: 72 Blackwell Ultra GPUs; rounded 20 TB GPU memory; 130 TB/s rack NVLink. No ATMOS access to that fabric is implied. Reviewed 5 October 2026.

S9. Tenstorrent: vLLM TT Plugin

<https://github.com/tenstorrent/vllm-tt-plugin>

Basis: A public Tenstorrent/vLLM integration path with model/device/version limits. Does not establish an ATMOS HBF backend or ATMOS performance. Reviewed 5 October 2026.

S10. LinkedIn authors: LiRank (2024; revised 7 Aug 2024)

<https://arxiv.org/abs/2402.06859>

Basis: Production-scale feed, jobs and ads ranking; quantization/vocabulary compression. Target-market precedent and incumbent optimization, not ATMOS validation. Reviewed 5 October 2026.

S11. Meta: Multi-Stage Architecture for Ads Ranking (5 Aug 2026)

<https://engineering.fb.com/2026/08/05/ml-applications/from-user-sequences-to-scaling-laws-a-multi-stage-architecture-for-metas-ads-ranking/>

Basis: Heavy offline user modeling and cached embeddings with latency-constrained online ranking. Co-design and caching must be included in the comparator. Reviewed 5 October 2026.

S12. Agent Router capabilities (formerly Envoy AI Gateway)

<https://theagentrouter.ai/docs/capabilities/>

Basis: Provider/pool routing, quota, rate limits and metrics. A candidate gateway foundation, not an ATMOS-ready distribution. Reviewed 5 October 2026.

S13. OpenAI: API pricing, specialized embedding models

<https://developers.openai.com/api/docs/pricing/>

Basis: Published standard input-token prices: text-embedding-3-small \$0.02 and text-embedding-3-large \$0.13 per million tokens. Policy/quality/SLO and non-token costs differ from private infrastructure. Reviewed 5 October 2026.